Uncertainty principles from majorization and entropy in the binary case

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Uncertainty relations quantify the impossibility of preparing a quantum system so that two incompatible measurements both have predictable outcomes. Majorization provides a particularly economical formulation: a single probability vector, independent of the state, bounds the outcome statistics of all states and thereby every reasonable measure of uncertainty at once, including the Shannon, Rényi and Tsallis entropies. Here we work out this approach completely in the simplest non-trivial setting, a pair of two-outcome measurements, and compare it systematically with uncertainty relations tailored to specific entropies. Using the elliptical geometry of the expectation values that qubit states can reach, we show that the known majorization bounds, of both direct-sum and tensor-product type, cannot be improved, yet no state attains them unless the measurements commute. Consequently, for non-commuting measurements the entropic bounds they imply are never tight for the Shannon entropy, the Tsallis entropies and the Rényi entropies of finite non-zero order; they become exact only at the two extremes of the Rényi family. We chart where each type of bound is preferable and how both compare with the Maassen–Uffink relation and with the exact optimum: the majorization bounds outperform Maassen–Uffink unless the measurements are close to mutually unbiased, and the direct-sum bound becomes tight to leading order as the measurements approach compatibility. Finally, we obtain the optimal bound for arbitrary noisy and biased two-outcome qubit measurements, show that the corresponding preparation uncertainty persists even when the two measurements are jointly measurable, and extend the results for projective measurements to binary measurements in any finite dimension.

I. INTRODUCTION

The uncertainty principle is one of the defining features of quantum mechanics [1–3]: there are pairs of measurements whose outcomes cannot both be predicted with certainty, however carefully the system is prepared. First formulated for the position and momentum of a particle, it has since become a quantitative tool of quantum information science. It underlies the security of quantum key distribution [4–6], provides experimentally accessible criteria for entanglement [5, 7, 8], and is closely related to, though distinct from, the incompatibility of quantum measurements [9, 10]. These applications call for formulations that apply to measurements with a finite number of outcomes and whose bounds do not depend on the state of the system.

Robertson's variance-based relation [2] does not meet these requirements: its lower bound depends on the state, it can vanish even for incompatible observables, and variances are not natural when the outcomes are mere labels [11]. Entropies of the outcome distributions provide a remedy [11–13]. The best known entropic uncertainty relation, due to Maassen and Uffink [14], bounds the sum of the Shannon entropies of the outcomes of two measurements by a quantity that depends only on the largest overlap between their eigenvectors. Entropic uncertainty relations have since grown into a broad field with applications throughout quantum information theory [15, 16]; they have been refined in various regimes [17, 18] and tested in experiments [19, 20].

Each such relation, however, constrains a single measure of uncertainty, while different tasks call for different measures: the Shannon entropy quantifies average information, the min-entropy governs guessing probabilities [21], and the Rényi and Tsallis families interpolate between these and other extremes. Majorization [22], the order that captures when one probability distribution is more disordered than another, makes it possible to treat all of them at once. It is a central tool of quantum information theory, where it governs, for instance, the transformations of entangled states [23] and, more generally, quantum resource theories [24, 25]. In the context of uncertainty, the idea is to find a single probability vector, independent of the state, that majorizes the outcome statistics of every state; every uncertainty measure that respects majorization—the Shannon, Rényi and Tsallis entropies among many others—then obeys a bound determined by this one vector. This programme was initiated by Partovi [26] and developed into universal uncertainty relations by Friedland, Gheorghiu and Gour [27] and by Puchała, Rudnicki and Życzkowski [28]; Rudnicki *et al.* subsequently introduced a direct-sum formulation that yields stronger entropic bounds [29]. Majorization uncertainty relations have since been extended to generalized measurements and quantum operations [30, 31],

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sharpened further [32, 33], in particular by exploiting the lattice structure of majorization, and verified experimentally [33].

The price of this universality is less clear. For a qubit measured in two bases—the setting of the BB84 protocol [4] and of many experiments—entropy-specific relations can be optimized exactly: the optimal Shannon relation is known [34, 35], and so are optimal relations for Rényi entropies [36, 37]. This raises a natural question: in the simplest setting, where both approaches can be carried out completely, what exactly does majorization gain, and what does it cost?

In this paper we answer this question for pairs of binary (two-outcome) measurements. For two qubit bases, the relevant majorization bounds are special cases of the general constructions of Refs. [27–29], and for unbiased unsharp qubit observables the direct-sum bound was evaluated in Ref. [31]. Building on these results, we give a unified and self-contained treatment of the binary case. We show that the expectation values that qubit states can reach fill an ellipse, and use this geometry to derive both bounds, to prove that they cannot be improved, and to identify the states that come closest to them. No single state attains these bounds unless the measurements commute. Consequently, for non-commuting measurements the entropic bounds they imply are strict for the Shannon entropy, for the Tsallis entropies and for all Rényi entropies except those of order zero and infinity. We then compare the two types of majorization bound with each other, with the Maassen–Uffink relation and with the exact optimum across the whole Rényi family. Finally, we derive the optimal bound for arbitrary noisy and biased two-outcome qubit measurements, discuss its relation to joint measurability, and extend the results for projective measurements to binary measurements in any finite dimension.

The paper is organized as follows. Section II introduces majorization, the uncertainty measures and the measurements we consider, and Sec. III describes the geometry of the achievable statistics. Section IV derives the optimal bounds for projective qubit measurements, and Sec. V discusses their entropic consequences and compares them with entropy-specific relations. Sections VI and VII treat general two-outcome qubit POVMs and binary projective measurements in arbitrary dimension, respectively, and Sec. VIII concludes. All results are verified numerically with the accompanying code (Appendix A).

II. PRELIMINARIES

A. Majorization and uncertainty measures

For $x \in \mathbb{R}^d$ let $x^\downarrow$ denote its entries in non-increasing order and $L_k(x) = \sum_{i=1}^{k} x_i^\downarrow$ its Lorenz partial sums. For x, y with equal total, x is *majorized* by y, written $x \prec y$, if $L_k(x) \le L_k(y)$ for all k [22]. A function Φ is Schur-concave if $x \prec y$ implies $\Phi(x) \ge \Phi(y)$, and *strictly* Schur-concave if, in addition, $\Phi(x) > \Phi(y)$ whenever $x \prec y$ and $x^\downarrow \neq y^\downarrow$. Every function $\Phi(x) = \sum_i \phi(x_i)$ with ϕ (strictly) concave is (strictly) Schur-concave [22].

The uncertainty measures used below are the Rényi [38] and Tsallis [39] entropies

$$H_\alpha(p) = \frac{\log_2 \sum_i p_i^\alpha}{1-\alpha}, \qquad T_q(p) = \frac{1 - \sum_i p_i^q}{q-1}, \tag{1}$$

with the Shannon entropy $H = H_1$ as the common limit ($T_q \to H \ln 2$ as $q \to 1$). The orders $\alpha = \infty$ and $\alpha = \frac{1}{2}$ give the min-entropy $H_\infty(p) = -\log_2 \max_i p_i$ and the max-entropy $H_{1/2}$, and H_0 is the logarithm of the number of non-zero entries. All of them are Schur-concave; H_α is strictly Schur-concave for $0 < \alpha < \infty$ and T_q for $q > 0$, whereas H_0 and H_∞ are not.

Let $p(\rho)$ and $q(\rho)$ denote the outcome distributions of two measurements M and N on a state ρ. For measurements in orthonormal bases $\{|m_i\rangle\}$ and $\{|n_j\rangle\}$, the Maassen–Uffink (MU) relation [14] reads

$$H(p) + H(q) \;\ge\; -\log_2 c, \qquad c = \max_{i,j} |\langle m_i | n_j \rangle|^2. \tag{2}$$

It constrains one particular functional of the pair (p, q); a different uncertainty measure requires a different relation. A majorization uncertainty relation [26–29] instead consists of a single state-independent vector ω such that

$$p(\rho) \oplus q(\rho) \;\prec\; \omega \quad \text{or} \quad p(\rho) \otimes q(\rho) \;\prec\; \omega \qquad \forall \rho. \tag{3}$$

We call these the direct-sum and the tensor-product form, respectively. Every Schur-concave function Φ then inherits the bound $\Phi(p \oplus q) \ge \Phi(\omega)$ or $\Phi(p \otimes q) \ge \Phi(\omega)$.

Restricted to non-increasing vectors of fixed total, $\prec$ is a partial order, and in fact a complete lattice [40, 41].

Definition 1. Let A be a set of vectors with common total. A vector ω is an *optimal majorization bound* for A if $a \prec \omega$ for all $a \in A$, and $\omega \prec \omega'$ for every other ω' with this property; i.e., $\omega = \sup A$ in the majorization lattice.

Lemma 1 ([40, 41])**.** *Let $S_k = \sup_{a \in A} L_k(a)$. Then $\sup A$ is the vector whose Lorenz curve is the least concave majorant of $k \mapsto S_k$ (with $S_0 = 0$). In particular, if $\omega_k = S_k - S_{k-1}$ is already non-increasing, then $\omega = \sup A$.*

In all cases treated below the second, simpler statement applies. The quantities S_k have a transparent meaning: S_k is the largest total probability that any state can concentrate on k outcomes.

B. Two-outcome qubit measurements

A qubit state is $\rho = \frac{1}{2}(\mathbb{1} + \vec{r} \cdot \vec{\sigma})$ with $|\vec{r}| \le 1$. A projective two-outcome measurement is $M = \{P_\pm\}$ with $P_\pm = \frac{1}{2}(\mathbb{1} \pm \hat{m} \cdot \vec{\sigma})$, $|\hat{m}| = 1$, and similarly $N = \{Q_\pm\}$ with axis $\hat{n}$. The outcome distributions are

$$p = \left(\tfrac{1+x}{2}, \tfrac{1-x}{2}\right), \; q = \left(\tfrac{1+y}{2}, \tfrac{1-y}{2}\right), \quad x = \hat{m} \cdot \vec{r}, \; y = \hat{n} \cdot \vec{r}. \tag{4}$$

The overlaps are $|\langle m_\pm|n_\pm\rangle|^2 = \frac{1}{2}(1 + \hat{m}\cdot\hat{n})$ and $|\langle m_\pm|n_\mp\rangle|^2 = \frac{1}{2}(1 - \hat{m}\cdot\hat{n})$, hence

$$c = \tfrac{1}{2}\big(1+|\hat{m}\cdot\hat{n}|\big) = \cos^2\tfrac{\theta}{2}, \qquad \cos\theta = |\hat{m}\cdot\hat{n}|,\ \theta\in[0,\tfrac{\pi}{2}]. \tag{5}$$

Relabelling the outcomes of N maps $\hat{n}\mapsto -\hat{n}$ and does not affect any majorization statement, so we may and shall assume $\hat{m}\cdot\hat{n} = \cos\theta \geq 0$. Note that $\frac{1}{2}\leq c\leq 1$ and $\sqrt{c}\geq 1/\sqrt{2} > 1-\sqrt{c}$.

A general two-outcome POVM $\{E_0, E_1 = \mathbb{1} - E_0\}$ can always be written as

$$E_0 = \tfrac{1}{2}\big[(1+\mu)\mathbb{1} + \eta\,\hat{a}\cdot\vec{\sigma}\big], \qquad \eta\geq 0,\ |\mu|+\eta\leq 1, \tag{6}$$

the constraint being equivalent to $0\leq E_0\leq\mathbb{1}$. Here η is the sharpness and μ the bias; rank-one projective measurements correspond to $\eta = 1$, $\mu = 0$.

III. GEOMETRY OF THE ACHIEVABLE SET

The statistics (4) depend on $\vec{r}$ only through (x, y).

Lemma 2. *For non-parallel axes, the set $\{(\hat{m}\cdot\vec{r}, \hat{n}\cdot\vec{r}) : |\vec{r}|\leq 1\}$ is the filled ellipse*

$$\mathcal{E}_\theta = \big\{(x,y) : x^2 - 2xy\cos\theta + y^2 \leq \sin^2\theta\big\}. \tag{7}$$

Its boundary consists exactly of the images of the pure states whose Bloch vectors lie in $\mathrm{span}\{\hat{m},\hat{n}\}$. *Moreover, for all $\alpha,\beta\in\mathbb{R}$,*

$$\max_{|\vec{r}|\leq 1}(\alpha x+\beta y) = |\alpha\hat{m}+\beta\hat{n}|. \tag{8}$$

Proof. Write $\vec{r} = \vec{r}_\parallel + \vec{r}_\perp$ with $\vec{r}_\parallel\in\mathrm{span}\{\hat{m},\hat{n}\}$. Only $\vec{r}_\parallel$ contributes to (x,y), and $(x,y) = G\xi$ where ξ are the coefficients of $\vec{r}_\parallel$ in the basis $\{\hat{m},\hat{n}\}$ and $G = \left(\begin{smallmatrix}1 & \cos\theta\\ \cos\theta & 1\end{smallmatrix}\right)$ is the Gram matrix. Then $|\vec{r}_\parallel|^2 = \xi^T G\xi = (x,y)G^{-1}(x,y)^T\leq 1$, which is the stated inequality since $\det G = \sin^2\theta$. Equality holds iff $\vec{r}_\perp = 0$ and $|\vec{r}| = 1$. Eq. (8) is Cauchy–Schwarz: $\alpha x+\beta y = \vec{r}\cdot(\alpha\hat{m}+\beta\hat{n})$. □

Consequently the achievable set $\wp(M,N) = \{p(\rho)\oplus q(\rho)\}\subset\mathbb{R}^4$ is the image of $\mathcal{E}_\theta$ under the affine map $(x,y)\mapsto\frac{1}{2}(1+x, 1-x, 1+y, 1-y)$: it is convex and compact, and its extreme points are the statistics of pure states in the plane of the two axes (Fig. 1). Any uncertainty measure that is concave in (x,y), such as $H(p)+H(q)$, is therefore minimized on $\partial\mathcal{E}_\theta$, and the optimization problems below are effectively one-dimensional.

IV. PROJECTIVE MEASUREMENTS

A. Direct-sum bound

Theorem 1. *Let M, N be projective qubit measurements with maximal overlap c. For every state ρ,*

$$p(\rho)\oplus q(\rho)\ \prec\ \omega_\oplus := \big(1,\ \sqrt{c},\ 1-\sqrt{c},\ 0\big), \tag{9}$$

and $\omega_\oplus$ is the optimal majorization bound for $\wp(M,N)$.

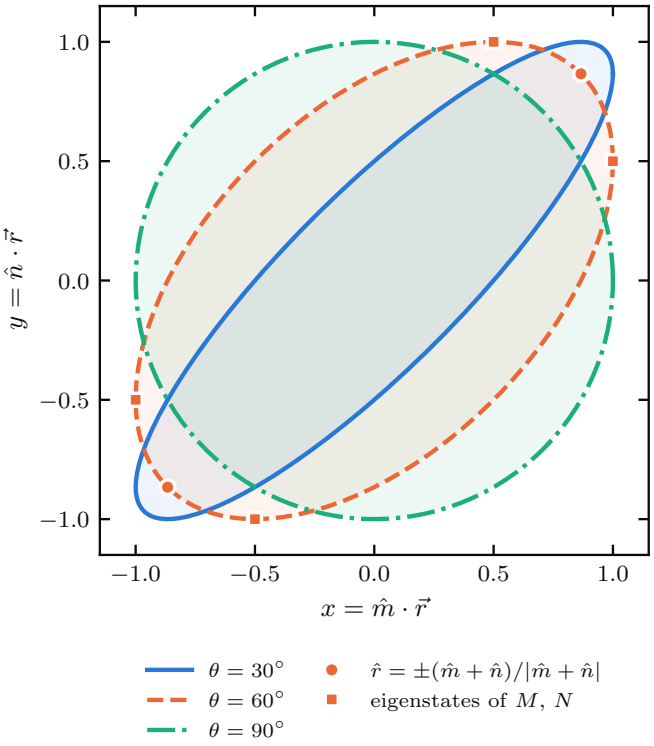


FIG. 1. The ellipse $\mathcal{E}_\theta$ of achievable expectation values $(x,y) = (\langle\hat{m}\cdot\vec{\sigma}\rangle, \langle\hat{n}\cdot\vec{\sigma}\rangle)$ for three Bloch angles. Markers (for $\theta = 60°$) show the two families of states that saturate the partial sums of $\omega_\oplus$: eigenstates of M or N (squares; $k = 1, 3$) and bisector states $\hat{r}\propto\pm(\hat{m}+\hat{n})$ (circles; $k = 2$). No single point saturates all partial sums (Proposition 1).

The majorization relation (9) is the qubit case of the direct-sum relation $p\oplus q\prec\{1\}\oplus W$ of Rudnicki *et al.* [29]: for $d = 2$ their vector $W = (s_1, s_2 - s_1)$ has $s_1 = \sqrt{c}$ and $s_2 = 1$. We give a short proof based on Lemma 2, which also yields optimality and the maximizing states.

Proof. We compute $S_k = \max_\rho L_k(p\oplus q)$ for $k = 1,\ldots,4$. Clearly $S_1\leq 1$ and $S_4 = 2$. Since all entries are non-negative, $S_3 = 2 - \min_\rho\min_i(p\oplus q)_i\leq 2$. Both bounds are attained by an eigenstate of M, for which $p = (1,0)$. For $k = 2$, the two largest entries either belong to the same distribution, with sum 1, or are of the form $p_i + q_j$. By Lemma 2,

$$p_i+q_j = 1+\tfrac{1}{2}(\pm x\pm y)\leq 1+\tfrac{1}{2}\max\{|\hat{m}+\hat{n}|, |\hat{m}-\hat{n}|\}. \tag{10}$$

With $\hat{m}\cdot\hat{n} = \cos\theta\geq 0$ the maximum is $|\hat{m}+\hat{n}| = \sqrt{2(1+\cos\theta)} = 2\cos\frac{\theta}{2} = 2\sqrt{c}$, so

$$S_2 = 1+\sqrt{c}, \tag{11}$$

attained by the bisector state $\vec{r} = (\hat{m}+\hat{n})/|\hat{m}+\hat{n}|$. This is the qubit case of the Landau–Pollak-type inequality $p_i + q_j\leq 1+\sqrt{c}$ underlying Deutsch's relation [11, 42]. The increments of $(S_1, S_2, S_3, S_4) = (1, 1+\sqrt{c}, 2, 2)$ give $\omega_\oplus$, which is non-increasing because $1\geq\sqrt{c}\geq 1-\sqrt{c}\geq 0$. Lemma 1 completes the proof. □

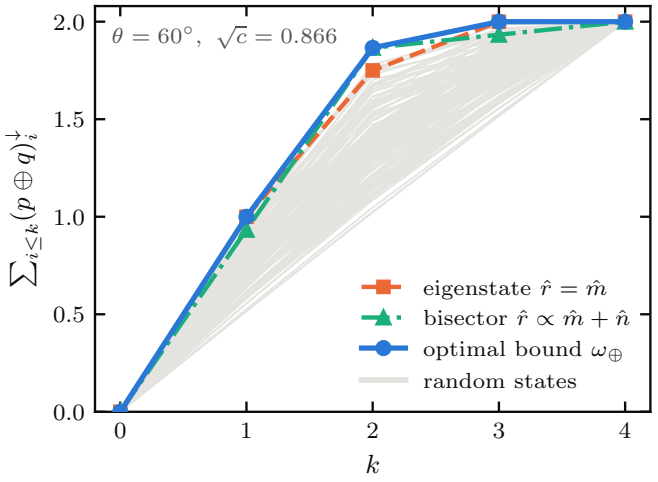


FIG. 2. Lorenz curves of $p \oplus q$ for $\theta = 60°$. Grey: 150 random states. The optimal bound $\omega_\oplus$ (blue) is the upper envelope; it touches the eigenstate curve at $k = 1, 3$ and the bisector-state curve at $k = 2$, but no single state reaches it.

Proposition 1. *There is a state with $(p \oplus q)^\downarrow = \omega_\oplus$ if and only if $c = 1$, i.e., M and N commute. For $c < 1$, the maxima S_1, S_3 are attained only by the eigenstates of M and N, and S_2 only by the bisector states $\vec{r} = \pm(\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$ (for $\theta = \pi/2$ also by $\vec{r} = \pm(\hat{m} - \hat{n})/|\hat{m} - \hat{n}|$).*

Proof. $(p \oplus q)^\downarrow_1 = 1$ forces ρ to be an eigenstate of M or N, say $\vec{r} = \pm\hat{m}$. Then $q = \left(\frac{1\pm\cos\theta}{2}, \frac{1\mp\cos\theta}{2}\right)$ and $(p \oplus q)^\downarrow = (1, c, 1-c, 0)$, which equals $\omega_\oplus$ iff $c = \sqrt{c}$, i.e., $c = 1$. The statement about S_2 follows from the equality condition in Cauchy–Schwarz and from $|\hat{m} - \hat{n}| < |\hat{m} + \hat{n}|$ for $\theta < \pi/2$. □

Proposition 1 makes precise the sense in which $\omega_\oplus$ is a *description* of the achievable set rather than the statistics of a particular state: it is the supremum of $\wp(M, N)$ in the majorization lattice, and its partial sums are saturated by two disjoint families of states (Fig. 2). The gap between $(1, c, 1-c, 0)$ and $\omega_\oplus$ quantifies the price one pays for demanding certainty about M.

B. Tensor-product bound

Theorem 2. *For every state ρ,*

$$p(\rho) \otimes q(\rho) \ \prec\ \omega_\otimes := \big(w,\ 1-w,\ 0,\ 0\big), \tag{12}$$

where $w = \frac{1}{4}\big(1 + \sqrt{c}\big)^2$, and $\omega_\otimes$ is the optimal majorization bound for $\{p(\rho) \otimes q(\rho)\}$.

The vector $\omega_\otimes$ coincides with the vector $Q^{(1)} = (C^2, 1-C^2, 0, 0)$, $C = (1+\sqrt{c})/2$, of Puchała *et al.* [28], which for $d = 2$ exhausts their hierarchy of bounds; see also Ref. [27].

Proof. By the arithmetic–geometric mean inequality and Eq. (11), $p_i q_j \le \left(\frac{p_i+q_j}{2}\right)^2 \le \left(\frac{1+\sqrt{c}}{2}\right)^2 = w$, with equality for the bisector state, for which $p_+ = q_+ = (1+\sqrt{c})/2$. Hence $S_1 = w$. For an eigenstate of M, $p \otimes q = (q_+, q_-, 0, 0)$, so $S_2 = S_3 = S_4 = 1$. Since $\sqrt{c} \ge 1/\sqrt{2}$ we have $w \ge \frac{1}{2}$, so the increments $(w, 1-w, 0, 0)$ are non-increasing, and Lemma 1 applies. □

As before, $\omega_\otimes$ is not attained for $c < 1$: equality in both steps of the proof forces a bisector state, at which $L_2(p \otimes q) = p_+ = (1+\sqrt{c})/2 < 1$.

V. ENTROPIC UNCERTAINTY RELATIONS

Let $h(t) = -t\log_2 t - (1-t)\log_2(1-t)$ be the binary entropy.

A. Bounds for the Rényi and Tsallis families

Corollary 1. *For every concave ϕ on $[0,1]$, $\sum_i \phi(p_i) + \sum_j \phi(q_j) \ge \phi(1) + \phi(\sqrt{c}) + \phi(1-\sqrt{c}) + \phi(0)$. In particular,*

$$H(p) + H(q) \ \ge\ h(\sqrt{c}), \tag{13}$$

$$T_q(p) + T_q(q) \ \ge\ \frac{1 - c^{q/2} - (1-\sqrt{c})^q}{q-1} \qquad (q > 0), \tag{14}$$

and, for $0 < \alpha \le 1$,

$$H_\alpha(p) + H_\alpha(q) \ \ge\ H_\alpha(\sqrt{c}, 1-\sqrt{c}), \tag{15}$$

where $H_\alpha(\sqrt{c}, 1-\sqrt{c}) = \log_2\big[c^{\alpha/2} + (1-\sqrt{c})^\alpha\big]/(1-\alpha)$.

Proof. The first statement is Karamata's inequality applied to Theorem 1. Eqs. (13) and (14) follow with $\phi(t) = -t\log_2 t$ and $\phi_q(t) = (t - t^q)/(q-1)$, since $\sum_i p_i = \sum_j q_j = 1$ gives $T_q(p) + T_q(q) = \sum_k \phi_q\big((p\oplus q)_k\big)$. Eq. (15) is the qubit case of Theorem 2 of Ref. [29]; we recall the short argument. For $\alpha < 1$ write $\sum_i p_i^\alpha = 1 + A$ and $\sum_j q_j^\alpha = 1 + B$ with $A, B \ge 0$. Then

$$\begin{aligned} H_\alpha(p) + H_\alpha(q) &= \frac{\log_2[(1+A)(1+B)]}{1-\alpha} \\ &\ge \frac{\log_2(1+A+B)}{1-\alpha}, \end{aligned}$$

and $1 + A + B = \sum_k (p\oplus q)_k^\alpha - 1 \ge \sum_k (\omega_\oplus)_k^\alpha - 1 = c^{\alpha/2} + (1-\sqrt{c})^\alpha$ by Schur-concavity of $v \mapsto \sum_k v_k^\alpha$. □

Corollary 2. *For every $\alpha \in [0, \infty]$,*

$$H_\alpha(p) + H_\alpha(q) \ \ge\ H_\alpha(\omega_\otimes) = \frac{\log_2\big[w^\alpha + (1-w)^\alpha\big]}{1-\alpha}. \tag{16}$$

For $\alpha = 1$ this reads $H(p) + H(q) \ge h(w)$; for $\alpha = \infty$ it gives $H_\infty(p) + H_\infty(q) \ge -2\log_2\frac{1+\sqrt{c}}{2}$, which is tight (bisector state); since $H \ge H_\infty$, it contains Deutsch's Shannon bound [11], which has the same right-hand side.

The two forms are natural for different families: Tsallis entropies are additive over direct sums, as used in the proof of Corollary 1, whereas Rényi entropies are additive over tensor products, $H_\alpha(p\otimes q) = H_\alpha(p) + H_\alpha(q)$. Where both forms apply, the direct sum is stronger, but it cannot be extended to all orders:

Proposition 2. *(i) For $0 < \alpha \le 1$, $H_\alpha(\sqrt{c}, 1-\sqrt{c}) \ge H_\alpha(\omega_\otimes)$, with equality iff $c = 1$; in particular $h(\sqrt{c}) \ge h(w)$. (ii) For $c < 1$ the direct-sum form (15) fails at $\alpha = \infty$: $\min_\rho [H_\infty(p) + H_\infty(q)] = -2\log_2 \frac{1+\sqrt{c}}{2} < -\frac{1}{2}\log_2 c = H_\infty(\sqrt{c}, 1-\sqrt{c})$.*

Proof. (i) Since $w - \sqrt{c} = \frac{1}{4}(1-\sqrt{c})^2 \ge 0$, we have $(\sqrt{c}, 1-\sqrt{c}) \prec (w, 1-w)$, with equality iff $c = 1$; apply strict Schur-concavity. The analogous statement in any dimension is proven in Ref. [29]. (ii) With $t = \sqrt{c}$, $\left(\frac{1+t}{2}\right)^2 - t = \left(\frac{1-t}{2}\right)^2 > 0$ for $t < 1$; the minimum is the tight bound of Corollary 2. □

B. When are the bounds tight?

Theorem 3. *Let $c < 1$. (i) For every continuous, strictly Schur-concave Φ,*

$$\min_\rho \Phi(p\oplus q) > \Phi(\omega_\oplus), \qquad \min_\rho \Phi(p\otimes q) > \Phi(\omega_\otimes). \quad (17)$$

Consequently the bounds (13), (14) and (15), and the bounds (16) with $0 < \alpha < \infty$, are all strict. (ii) The bounds (16) of order $\alpha = 0$ and $\alpha = \infty$ are tight; they are attained by the eigenstates of M or N and by the bisector states, respectively.

Proof. (i) The achievable statistics form a compact set, so the minimum of Φ is attained at some state ρ^*. By Theorem 1 and Proposition 1, $p\oplus q(\rho^*) \prec \omega_\oplus$ with $(p\oplus q)^\downarrow \ne \omega_\oplus$, hence $\Phi(p\oplus q(\rho^*)) > \Phi(\omega_\oplus)$. The tensor product is analogous, using the non-attainment of $\omega_\otimes$ shown after Theorem 2. The Shannon and Tsallis sums are strictly Schur-concave functions of $p\oplus q$, and $H_\alpha(p) + H_\alpha(q) = H_\alpha(p\otimes q)$ with H_α strictly Schur-concave for $0 < \alpha < \infty$. For (15) with $\alpha < 1$, apply the argument to $v \mapsto \sum_k v_k^\alpha$ in the proof of Corollary 1. (ii) An eigenstate of M gives $H_0(p) + H_0(q) = 0 + 1 = H_0(\omega_\otimes)$, and a bisector state gives $\max_i p_i \max_j q_j = w$. □

Theorem 3 makes precise in what sense the majorization bounds are optimal. As majorization bounds they cannot be improved (Theorems 1 and 2); yet, because the supremum is not the statistics of any state, the bound it induces is strict for every strictly Schur-concave measure. For such measures the optimum has to be found measure by measure, which is what entropy-specific relations do; only at the extreme orders $\alpha = 0$ and $\alpha = \infty$ does the majorization bound reach it.

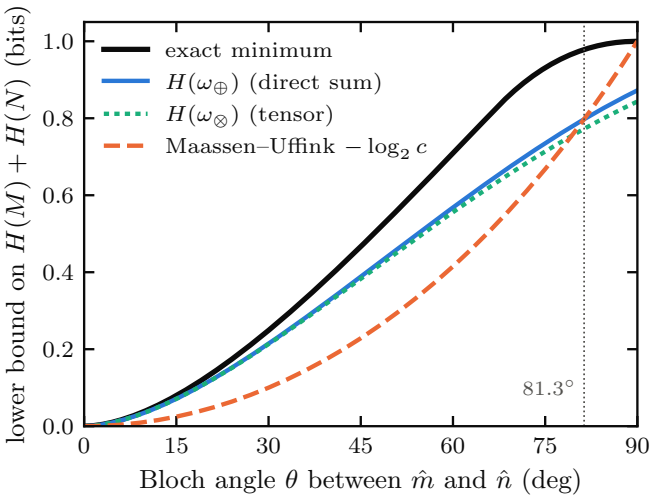


FIG. 3. Lower bounds on $H(M) + H(N)$ as a function of the Bloch angle θ ($c = \cos^2\frac{\theta}{2}$). Black: exact minimum, equal to $2h\left(\frac{1+\sqrt{c}}{2}\right)$ for $\theta \le \bar{\theta} \approx 67.07°$. Blue: direct-sum bound $h(\sqrt{c})$. Green (dotted): tensor-product bound $h(w)$. Orange (dashed): Maassen–Uffink. The direct-sum bound is stronger than MU for $\theta < 81.3°$ (dotted vertical line).

C. Comparison with the Maassen–Uffink relation and with the exact optimum

Figure 3 compares the Shannon bounds with MU and with the exact minimum of $H(p) + H(q)$, which is known for qubits [34, 35]. By Lemma 2 the minimum lies on the great circle through $\hat{m}$ and $\hat{n}$. For $\theta \le \bar{\theta}$ it is attained at the bisector states and equals $2h\left(\frac{1+\sqrt{c}}{2}\right)$ [35]. At $\bar{\theta}$ the bisector turns from a minimum into a saddle point along the circle; a second-derivative test shows that this happens when $\sqrt{\bar{c}}\,\mathrm{artanh}\,\sqrt{\bar{c}} = 1$, i.e., at $\bar{c} \approx 0.6948$ and $\bar{\theta} \approx 67.07°$. For $\bar{\theta} < \theta < \pi/2$ the minimizers bifurcate and the minimum has to be computed numerically [35].

For mutually unbiased measurements ($\theta = 90°$, $c = \frac{1}{2}$) MU is tight (1 bit), while $h(1/\sqrt{2}) \approx 0.872$ and $h(w) \approx 0.844$. As the measurements become more compatible, however, the majorization bounds take over: $h(\sqrt{c}) > -\log_2 c$ for $\sqrt{c} > t^* \approx 0.7585$, i.e., for Bloch angles $\theta < 81.3°$ (the tensor bound overtakes MU at $\theta < 79.3°$), in agreement with Ref. [31].

As $c \to 1$ the advantage over MU becomes unbounded, and the direct-sum bound becomes tight in ratio. With $\varepsilon = 1 - \sqrt{c} \approx \theta^2/8$ one has $-\log_2 c \approx 2\varepsilon/\ln 2$ and $h(\sqrt{c}) = h(\varepsilon) \approx \varepsilon\log_2(e/\varepsilon)$, whereas the exact minimum is $2h(\varepsilon/2) \approx \varepsilon\log_2(2e/\varepsilon)$. To leading order the ratio of the exact minimum to $h(\sqrt{c})$ is therefore $1 + 1/\log_2(e/\varepsilon)$: it tends to 1, but only logarithmically (1.09 at $\theta = 0.1$, 1.047 at $\theta = 0.003$), while the ratio to MU grows as $\frac{1}{2}\ln(16e/\theta^2)$ (4.2 and 7.7). The eigenstates of M, although they saturate S_1 and S_3, give $h(c)$, asymptotically twice the minimum.

Figure 4 extends the comparison to the whole Rényi family at $\theta = 60°$ and $\theta = 85°$. For $\alpha \le 1$ MU also

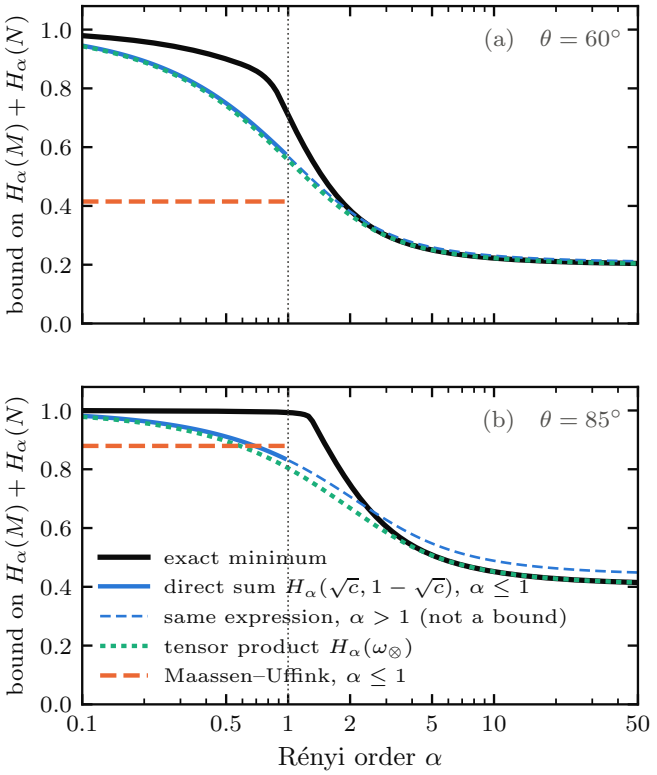


FIG. 4. Bounds on $H_\alpha(M) + H_\alpha(N)$ across the Rényi family for (a) $\theta = 60°$ and (b) $\theta = 85°$. Black: exact minimum. Blue: direct-sum bound $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ of Corollary 1, valid for $\alpha \le 1$; its thin dashed continuation for $\alpha > 1$ is not a bound and eventually exceeds the exact minimum. Green (dotted): tensor-product bound $H_\alpha(\omega_\otimes)$ of Corollary 2. Orange (dashed): MU, valid for $\alpha \le 1$. The dotted vertical line marks the Shannon case $\alpha = 1$.

applies, because H_α is non-increasing in α. At $\theta = 60°$ both majorization bounds beat MU for all $\alpha \le 1$; at $\theta = 85°$, closer to mutual unbiasedness, they do so only for $\alpha < 0.69$ (direct sum) and $\alpha < 0.58$ (tensor product). For the max-entropy, $\alpha = \frac{1}{2}$, the direct-sum bound reaches 83% and 91% of the exact minimum at the two angles. The expression $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ remains a valid bound somewhat beyond $\alpha = 1$ in these examples, but it fails above $\alpha \approx 2.1$ ($\theta = 60°$) and $\alpha \approx 2.5$ ($\theta = 85°$), in line with Proposition 2(ii). For large α only the tensor-product bound remains, and it converges to the exact minimum as $\alpha \to \infty$. The majorization bounds thus become exact only at the two ends of the family, as Theorem 3 requires.

Finally, MU is one member of a family $H_\alpha(p)+H_\beta(q) \ge -\log_2 c$ with $1/\alpha+1/\beta = 2$ [14], which includes the pair $H_\infty(p) + H_{1/2}(q) \ge -\log_2 c$ of min- and max-entropy. Majorization relations of the form (3) are symmetric under the exchange of p and q and therefore yield such mixed-order relations only indirectly, through the monotonicity of H_α in α.

VI. GENERAL TWO-OUTCOME QUBIT POVMS

For POVMs, the partial sums of the direct-sum relation are largest eigenvalues of sums of effects, $S_k = \max_{|J|=k} \lambda_{\max}\big(\sum_{j\in J} F_j\big)$, where the F_j run over the effects of both measurements [31] (see also Refs. [29, 30, 43]); Baek *et al.* evaluated them for unbiased unsharp qubit observables [31]. For two-outcome qubit POVMs the maximization can be carried out in closed form for arbitrary bias, and the resulting vector turns out to be ordered, hence optimal.

Let $M = \{E_0, E_1\}$ with parameters $(\mu, \eta, \hat{a})$ and $N = \{F_0, F_1\}$ with $(\nu, \zeta, \hat{b})$, as in Eq. (6). Then $p = \frac{1}{2}\big(1+\mu+\eta\,\hat{a}\cdot\vec{r},\ 1-\mu-\eta\,\hat{a}\cdot\vec{r}\big)$, and similarly for q.

Theorem 4. *Define*

$$\kappa = \max\{|\mu| + \eta,\ |\nu| + \zeta\}, \tag{18}$$

$$\Lambda = \max_{s,t=\pm1} \big[s\mu + t\nu + |s\eta\,\hat{a} + t\zeta\,\hat{b}|\big]. \tag{19}$$

Then $p(\rho) \oplus q(\rho) \prec \omega_\oplus$ for all ρ, where

$$\omega_\oplus = \tfrac{1}{2}\big(1+\kappa,\ 1+\Lambda-\kappa,\ 1+\kappa-\Lambda,\ 1-\kappa\big) \tag{20}$$

is the optimal majorization bound.

Proof. The largest single outcome probability is $\max_\rho \frac{1}{2}\big(1 + |\mu + \eta\,\hat{a}\cdot\vec{r}|\big) = \frac{1}{2}(1 + |\mu| + \eta)$ for M, so $S_1 = \frac{1}{2}(1+\kappa)$; likewise the smallest is $\frac{1}{2}(1-\kappa)$, so $S_3 = 2 - \frac{1}{2}(1-\kappa)$. For S_2, a pair within one distribution sums to 1, while a cross pair gives, by Eq. (8),

$$\max_\rho \big[1 + \tfrac{1}{2}\big(s(\mu + \eta\,\hat{a}\cdot\vec{r}) + t(\nu + \zeta\,\hat{b}\cdot\vec{r})\big)\big] = 1 + \tfrac{1}{2}\big[s\mu + t\nu + |s\eta\hat{a} + t\zeta\hat{b}|\big]. \tag{21}$$

Choosing $s = \operatorname{sgn}\mu$, $t = \operatorname{sgn}\nu$ shows $\Lambda \ge 0$, so $S_2 = 1 + \frac{1}{2}\Lambda$. The increments of (S_k) are the entries of Eq. (20). They are non-increasing iff $\kappa \le \Lambda \le 2\kappa$. The upper inequality follows from the triangle inequality, $\Lambda \le |\mu| + |\nu| + \eta + \zeta \le 2\kappa$. For the lower one, let ρ^* attain S_1 via an outcome of, say, M; then $S_2 \ge S_1 + \max_j q_j(\rho^*) \ge S_1 + \frac{1}{2}$, i.e., $\Lambda \ge \kappa$. Lemma 1 completes the proof. □

For projective measurements $\kappa = 1$, $\Lambda = 2\sqrt{c}$ and Eq. (20) reduces to Eq. (9). For unbiased measurements ($\mu = \nu = 0$), $\Lambda = (\eta^2 + \zeta^2 + 2\eta\zeta|\hat{a}\cdot\hat{b}|)^{1/2}$, in agreement with Ref. [31]. If moreover $\eta = \zeta$, then $\omega_\oplus = \eta\,(1, \sqrt{c}, 1-\sqrt{c}, 0) + \frac{1-\eta}{2}(1,1,1,1)$: such measurements are noisy versions of sharp ones [31, 44], and the achievable statistics, their supremum and the saturating states are those of the sharp case mixed with the uniform vector. For unequal sharpness or nonzero bias, $\omega_\oplus$ is not of this form, and the relevant geometric quantities are the lengths $|\eta\hat{a} \pm \zeta\hat{b}|$ rather than the overlap c.

Uncertainty versus incompatibility.—For unbiased POVMs the vectors $\eta\hat{a} \pm \zeta\hat{b}$ that determine Λ also decide joint measurability: the pair is jointly measurable

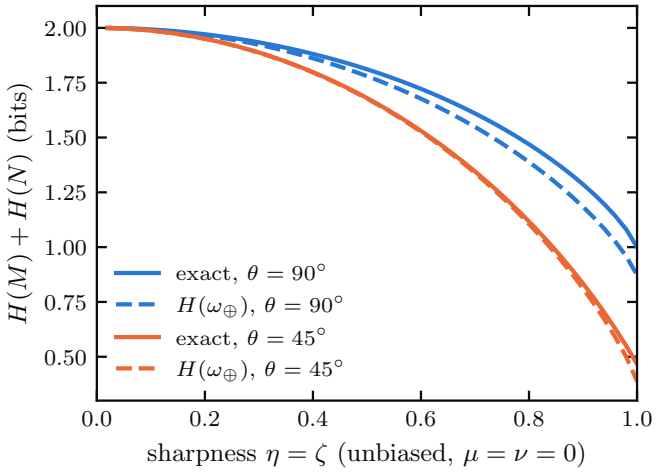


FIG. 5. Unsharp measurements ($\mu = \nu = 0$, $\eta = \zeta$): exact minimum of $H(M) + H(N)$ (solid) and the bound from Theorem 4 (dashed) for Bloch angles 90° and 45°.

if and only if $|\eta\hat{a} + \zeta\hat{b}| + |\eta\hat{a} - \zeta\hat{b}| \leq 2$ [9, 44]. The two notions are nevertheless distinct. For orthogonal axes and $\eta = \zeta \leq 1/\sqrt{2}$ the measurements are jointly measurable, yet $S_2 = 1 + \eta/\sqrt{2}$ stays below the value $1 + \eta$ that would be reached if both outcome distributions could be sharpened at once, and $\omega_\oplus$ is not attained, being a noisy version of the sharp bound (Proposition 1). Majorization uncertainty relations therefore quantify preparation uncertainty, which persists in the absence of measurement incompatibility.

As a corollary, $H(p) + H(q) \geq -\sum_k \omega_k \log_2 \omega_k$ for all two-outcome qubit POVMs. Figure 5 shows that this bound tracks the exact minimum over the whole range of sharpness, becoming exact in the trivial limit $\eta \to 0$.

VII. BINARY PROJECTIVE MEASUREMENTS IN DIMENSION d

The qubit analysis already contains the general binary projective case.

Theorem 5. *Let $M = \{P, \mathbb{1} - P\}$ and $N = \{Q, \mathbb{1} - Q\}$ be projective measurements on $\mathbb{C}^d$ and $c = \max_{i,j} \|P_i Q_j\|_\infty^2$. Then Theorems 1 and 2 hold verbatim, and $\omega_\oplus$ is attained if and only if $c = 1$.*

Proof. By Jordan's lemma [45], which is also the standard tool for entropic uncertainty relations of binary measurements [46], $\mathbb{C}^d$ decomposes into mutually orthogonal subspaces $\mathcal{H}_b$ of dimension one or two, each invariant under P and Q. On a two-dimensional block, P and Q act as rank-one projectors, i.e., as a pair of qubit measurements with overlap c_b; on a one-dimensional block they have a common eigenvector, and we set $c_b = 1$. Since P and Q are block diagonal, $p(\rho) \oplus q(\rho) = \sum_b \lambda_b\, p_b \oplus q_b$ with $\lambda_b = \mathrm{tr}(\rho\Pi_b)$ and p_b, q_b the statistics of the normalized block state $\Pi_b \rho \Pi_b / \lambda_b$. Each $p_b \oplus q_b \prec \omega_\oplus(c_b) \prec \omega_\oplus(c)$, because $\omega_\oplus$ increases in the majorization order with c and $c = \max_b c_b$. Since $\{v : v \prec \omega\}$ is convex, $p \oplus q \prec \omega_\oplus(c)$; optimality follows by embedding the maximizing qubit states of a block with $c_b = c$. For the tensor product, $p_i q_j \leq (\frac{p_i + q_j}{2})^2$ with $p_i + q_j \leq \lambda_{\max}(P_i + Q_j) = 1 + \|P_i Q_j\| \leq 1 + \sqrt{c}$; $S_1 = w$ is attained in a block with $c_b = c$, and $S_2 = 1$ by any state in the range of P. Finally, if $c < 1$, a state with $L_1(p \oplus q) = 1$ lies in the range of some P_i (or Q_j); then $q_j = \mathrm{tr}(\rho P_i Q_j P_i) \leq \|P_i Q_j\|^2 \leq c$ for all j, so $L_2(p \oplus q) \leq 1 + c < 1 + \sqrt{c}$ and $\omega_\oplus$ is not attained. If $c = 1$, the ranges of some P_i and Q_j intersect and a common eigenvector attains $\omega_\oplus = (1, 1, 0, 0)$. □

In particular, all bounds of Sec. V hold for binary projective measurements in any dimension with $c = \max_{i,j} \|P_i Q_j\|^2$, and the min-entropy bound remains tight. For arbitrary POVMs in any dimension, the eigenvalue formula for S_k quoted in Sec. VI, together with Lemma 1, gives an exact numerical procedure, which we use to verify Theorems 4 and 5 independently (Appendix A).

VIII. DISCUSSION

For binary measurements the majorization approach to uncertainty can be carried out completely. The achievable statistics form an affine image of an ellipse, and a single four-component vector, determined by the overlap c (or, for POVMs, by κ and Λ), is their least upper bound in the majorization lattice, for the direct sum as well as for the tensor product. This is the unified perspective that majorization offers: one vector controls every Schur-concave uncertainty measure at once.

The comparison with entropy-specific relations shows what this unification costs. Because the supremum is not the statistics of any state when the measurements do not commute, the induced bound is strict for every strictly Schur-concave measure, and it becomes exact only at the extreme orders $\alpha = 0$ and $\alpha = \infty$. Against a generic scalar relation such as MU, on the other hand, the majorization bounds are strong: the direct-sum Shannon bound beats MU for Bloch angles below 81.3° and is asymptotically tight in ratio as the measurements approach compatibility, precisely where MU is weakest. Among the two forms, the direct sum is preferable for $\alpha \leq 1$, while only the tensor product survives for large α.

Several questions remain open. (i) *Tensor bounds for POVMs.* The AM–GM step of Theorem 2 is no longer tight for biased or unequally sharp measurements; the optimal $\omega_\otimes$ then requires maximizing products of affine functions over $\mathcal{E}_\theta$, and the least concave majorant in Lemma 1 may become non-trivial. (ii) *Binary POVMs in dimension d.* Jordan's lemma does not apply to pairs of effects directly. A Naimark dilation [47] turns them into projectors, but only product inputs are allowed, so Theorem 5 does not apply as it stands. (iii) *More mea-*

surements. For three or more binary measurements the partial sums involve signed sums of several Bloch vectors, and the lattice structure of the bound becomes genuinely relevant. (iv) *Mixed orders.* It remains open whether majorization can be adapted to yield relations such as $H_\infty(p) + H_{1/2}(q) \geq -\log_2 c$.

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Appendix A: Numerical methods and code

All results were checked with the accompanying Python code (`code/`):

- `majorization.py`: POVM parametrization, Lorenz curves, the lattice supremum via least concave majorants [40], the analytic bounds of Theorems 1, 2 and 4 and Corollaries 1–2, the eigenvalue characterization $S_k = \max_{|J|=k} \lambda_{\max}(\sum_{j\in J} F_j)$, and exact minima of Rényi and Tsallis entropy sums.

- `verify_bounds.py`: compares the closed forms with (a) lattice suprema over 1.8×10^5 grid states (maximal deviation $< 10^{-4}$), and (b) the eigenvalue characterization for 200 random qubit POVM pairs and for 150 random pairs of projectors of rank $d/2$ in dimensions $d = 2, 4, 6$, all with $c < 1$ (deviation $< 10^{-14}$). Pairs with a common eigenvector, which always occur for odd d or unequal ranks, have $c = 1$ and are checked separately. The script also checks Propositions 1 and 2, Corollaries 1–2, Theorem 3, the critical angle $\bar{\theta}$, and the jointly measurable example of Sec. VI.

- `make_figures.py`: generates Figs. 1–5.

The exact entropy minima are obtained by a dense grid over the great circle in $\mathrm{span}\{\hat{m}, \hat{n}\}$ refined by Nelder–Mead. This reduction is justified by Lemma 2: the Shannon entropy is concave in (x, y) and is therefore minimized on the boundary of $\mathcal{E}_\theta$. For unbiased measurements, moving $\vec{r}$ outwards makes both p and q more ordered, so the same holds for every Rényi and Tsallis entropy.

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